



EUROPEAN COMMISSION 2026

# Foundations of Difference-in-Differences

*Day 1: The  $2 \times 2$ , Parallel Trends, and Event Studies*

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CUNEF Auditorium · June 3–4, 2026

# Who am I?

**Scott Cunningham** · Professor of Economics, Baylor University

- Empirical work: sex work, relationship markets, drug policy, abortion, corrections, suicide and mental illness — and AI agents and AI methodologies
- Podcast: *The Mixtape with Scott*
- Substack: *Scott's Substack*
- Teaching platform: *Mixtape Sessions*

## Day 1

### 9:00am – 2:30pm DiD intro and covariates

- Canonical  $2 \times 2$  difference-in-differences
- Conditional trends as functions of  $X_i$
- Estimation under conditional parallel trends: regression adjustment, IPW, doubly-robust
- Sub-group heterogeneous treatment effects
- Coding lab: covariates

### 2:30pm – 5:00pm Continuous DiD and coding lab

- DiD where the treatment is a non-negative “dose” (continuous DiD)
- Treatment effects under a dose: total effect and per-unit effect
- Interpretation difficulties: the dose-specific ATT
- Estimation of continuous DiD in  $2 \times 2$  settings
- Coding lab: continuous DiD

## Day 2

### 9:00am – 2:00pm Staggered treatment timing

- Staggered timing and the definition of effects:  $ATT(g, t)$
- Aggregation of group-time treatment effects
- Callaway and Sant'Anna estimator for  $ATT(g, t)$
- Imputation-based approaches

### 2:00pm – 5:00pm Claude Code and DiD

- Claude Code introduction and Skills (but it seems down for me)
- Claude Code DiD Skills and example

## Today: the basics of DiD

Four averages and three subtractions.

**Calculation.**

**Identification.**

**Regressions.**

**Falsifications.**

## Today: the basics of DiD

Covariates

**TWFE**

**Outcome regression**

**Propensity scores**

**Doubly robust**

## Continuous treatments

**TWFE**

**Target parameters**

**Identification**

**Estimation**

## Staggered adoption

**Various estimators**

**Target parameters**

**Identification**

**Estimation**



## Synthetic control methods

**Abadie's original**

**Bias adjustment for poor fit**

**Synthetic diff-in-diff**

Tomorrow

Claude Code

**To be determined due to some outage (for me)**

LESSON

# 1. DiD's Empirical Roots

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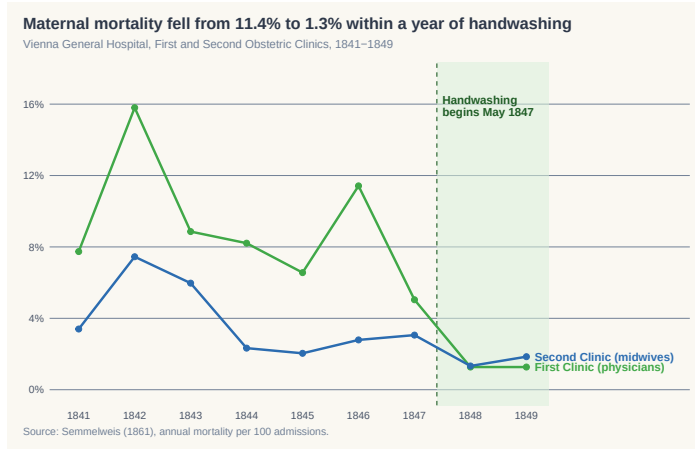
## Vienna, 1840s: two clinics, two mortality rates

- Vienna General Hospital, First Obstetric Clinic
- Pregnant women were assigned to the *physicians' wing* or the *midwives' wing* by day of admission — effectively at random
- Physicians' wing: **13–18% mortality** from puerperal fever
- Midwives' wing: **about 3%**
- Women would deliver in the street rather than risk the physicians' wing

## His hypothesis, and his DiD

- Semmelweis — an assistant on the wards — noticed the physicians did cadaver dissections before delivering babies. The midwives did not.
- Hypothesis: something on the physicians' hands was causing the fever.
- May 1847: he instituted **chlorinated lime handwashing for physicians only**
- Midwives served as the untreated comparison group — a DiD by hand

# The data



*Monthly mortality, First Obstetric Clinic. Handwashing began mid-1847.*

## Princeton, 1970s: Orley Ashenfelter

The Industrial Relations Section was founded 1922 — rigorous empirical labor research, the policy edge of US economics.

### Orley Ashenfelter's leadership

- Long-time director. Set the tone: high-quality data, careful design, communicate to bureaucrats.
- Trained a cohort: **David Card**, **Josh Angrist**, **Bob LaLonde**, **Janet Currie**, Pischke, Dinardo, Case, others.
- Hired **Alan Krueger** as a junior colleague.

## By the late 1970s, empirical labor was in trouble

- Different methods on the same question disagreed wildly
- The Princeton response: **design over modeling — estimators that mimic an RCT as closely as possible**
- Worker surveys, federal job training, massive panel regressions
- Audience was bureaucrats

*Their eyes would glaze over whenever you said ‘regression.’*

— Orley Ashenfelter

[youtu.be/WnB3EJ8K7lg](https://youtu.be/WnB3EJ8K7lg) (3:53)



LESSON

## 2. DiD is Two Things: a Calculation and an Assumption

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## DiD is a research design, not a regression

*A research design is an identification strategy: it reveals the invisible target parameter hidden inside a calculation.*

**1.**

***A calculation***

(the  $2 \times 2$ )

+

**2.**

***An assumption***

(parallel trends)

## The individual treatment effect

$$\delta_i = Y_i(1) - Y_i(0)$$

- One per unit. **Different** across units in general — treatment effects are heterogeneous.
- We only ever see one of  $Y_i(1)$  or  $Y_i(0)$ . The other is the counterfactual.
- So  $\delta_i$  is never directly observed at the individual level.

## The ATT: a causal parameter has three ingredients

$$\text{ATT} = \underbrace{\mathbb{E}}_{\text{weighted expectations}} \left[ \underbrace{Y(1) - Y(0)}_{\text{potential outcomes}} \mid \underbrace{D = 1, \text{Post}}_{\text{population}} \right]$$

- **Potential outcomes** — the contrast  $Y(1) - Y(0)$
- **Population** — on whom: treated units, post-period
- **Weighted expectations** — how we average across that population ( $\mathbb{E}[\cdot]$  defaults to equal weights)

## The switching equation: you see one half

$$Y_i = D_i \cdot Y_i(1) + (1 - D_i) \cdot Y_i(0)$$

- For each unit:  $Y_i(1)$  if treated,  $Y_i(0)$  if not.
- The ATT asks about *both* columns. The data hands you one.

*The rest of the day: strategies for recovering the ATT from the half you see.*

LESSON

# 3. Five Regressions, One Number

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## Cheng & Hoekstra (2013): castle doctrine and homicide

**The reform.** “Castle doctrine” laws expanded the legal right to use lethal force in self-defense, removing the duty to retreat. Many US states passed expansions from **2005–2010**.

- **Panel:** all US states  $\times$  years (state-year).
- **Outcome:** log homicide rate, FBI Supplementary Homicide Reports.
- **Treatment:** state passed an expansion of castle doctrine.
- **Comparison:** states that did not pass (never-treated).

For today: a clean  $2 \times 2$  on **2005–2006**. Treated = states adopting in 2006; control = states that hadn't. *Cheng & Hoekstra (2013), “Does Strengthening Self-Defense Law Deter Crime or Escalate Violence? Evidence from Expansions to Castle Doctrine.” J. Human Resources 48(3): 821–854.*

## The four cells, by hand on castle.dta

```
use castle.dta, clear
keep if year >= 2005 & year <= 2006
gen post = (year == 2006)
gen treat = (effyear == 2006)

summarize l_homicide if treat==1 & post==1 // Y11
summarize l_homicide if treat==1 & post==0 // Y10
summarize l_homicide if treat==0 & post==1 // Y01
summarize l_homicide if treat==0 & post==0 // Y00
```

$$\widehat{\text{DiD}} = (\bar{Y}_{11} - \bar{Y}_{10}) - (\bar{Y}_{01} - \bar{Y}_{00}) \approx \mathbf{0.10}$$

*Log homicide rate ~ 10% higher in 2006-adopting states relative to non-adopters.*



## Regression 1: OLS with $\text{treat} \times \text{post}$

```
reg l_homicide post##treat, cluster(sid)
```

The coefficient on `1.post#1.treat` is the DiD.

$$Y = \beta_0 + \beta_1 \text{Post} + \beta_2 \text{Treat} + \beta_3 (\text{Post} \times \text{Treat}) + \varepsilon$$

*The saturated specification.  $\hat{\beta}_3 \approx 0.10$ .*

## Regression 2: two-way fixed effects

```
xtset sid year  
xtreg l_homicide c.treat#c.post i.year, fe vce(cluster sid)
```

State fixed effects absorb time-invariant level differences. Year fixed effects absorb common shocks.

*In the  $2 \times 2$  case, the TWFE coefficient equals the DiD. Same number.*

### Regression 3: first-difference $\Delta Y_i$ on $D_i$

```
keep sid year l_homicide treat  
reshape wide l_homicide, i(sid) j(year)  
gen dY = l_homicide2006 - l_homicide2005  
reg dY treat, vce(cluster sid)
```

$$\Delta Y_i = \alpha + \beta_{DD} D_i + u_i$$

- Collapse the panel: one  $\Delta Y$  per unit.
- Time-differencing wipes out unit fixed effects mechanically.
- Slope on treat = the DiD. Same number.

## Regression 4: $(Y_{\text{treat}} - Y_{\text{ctrl}})$ on the post dummy

At each time  $t$ , take the across-group difference  $D_t = \bar{Y}_{1,t} - \bar{Y}_{0,t}$ . Then regress on a post indicator:

$$D_t = \alpha + \beta_{DD} \text{Post}_t + v_t$$

```
collapse (mean) l_homicide, by(treat year)
reshape wide l_homicide, i(year) j(treat)
gen Dt = l_homicide1 - l_homicide0
gen post = (year == 2006)
reg Dt post
```

- Regression 3 differences over *time*; Regression 4 differences over *groups* first.
- By Frisch–Waugh–Lovell, partialling out the two-way means runs in either order. Same number.

## Regression 5: long differences — the Snow move

Compute one number per group, post minus pre, then subtract:

$$\widehat{\text{DiD}} = (\bar{Y}_{11} - \bar{Y}_{10}) - (\bar{Y}_{01} - \bar{Y}_{00})$$

- No regression. Two long differences. One subtraction between them.
- This is what John Snow did in 1854 with the water-company data.
- Identical to Regressions 1–4. **The arithmetic was the point all along.**

## Five regressions, one number

| Specification                                                 | What it does                                      | $\hat{\beta}_{DD}$ |
|---------------------------------------------------------------|---------------------------------------------------|--------------------|
| 1. OLS with <code>post##treat</code>                          | saturated interaction                             | $\approx 0.10$     |
| 2. Two-way fixed effects                                      | absorb unit + time means                          | $\approx 0.10$     |
| 3. $\Delta Y_i$ on $D_i$ (first difference)                   | difference over time, then on $D$                 | $\approx 0.10$     |
| 4. $D_t$ on <code>Post<sub>t</sub></code> (across-group diff) | difference over groups, then on <code>Post</code> | $\approx 0.10$     |
| 5. Long differences (Snow)                                    | two means, one subtraction                        | $\approx 0.10$     |

*Same data, five specifications, identical to machine epsilon.  
The  $2 \times 2$  is the object; the regressions are just five ways to ask for it.*

LESSON

# 4. Decomposing the $2 \times 2$

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## The proof: add a zero, rearrange (five steps)

### 1. Start with the $2 \times 2$ :

$$\widehat{\text{DiD}} = [\mathbb{E}[Y \mid 1, \text{Post}] - \mathbb{E}[Y \mid 1, \text{Pre}]] - [\mathbb{E}[Y \mid 0, \text{Post}] - \mathbb{E}[Y \mid 0, \text{Pre}]]$$

### 2. Switching equation. Treated-post is $Y(1)$ ; the other three cells are $Y(0)$ :

$$\widehat{\text{DiD}} = [\mathbb{E}[Y(1) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Pre}]] - [\mathbb{E}[Y(0) \mid 0, \text{Post}] - \mathbb{E}[Y(0) \mid 0, \text{Pre}]]$$

### 3. Add a zero. Insert $+\mathbb{E}[Y(0) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Post}]$ :

$$\begin{aligned}\widehat{\text{DiD}} &= \mathbb{E}[Y(1) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Post}] \\ &\quad + \mathbb{E}[Y(0) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Pre}] - [\mathbb{E}[Y(0) \mid 0, \text{Post}] - \mathbb{E}[Y(0) \mid 0, \text{Pre}]]\end{aligned}$$

### 4. Rearrange. First two terms = ATT; remainder = difference in $Y(0)$ trends:

$$\widehat{\text{DiD}} = \underbrace{\mathbb{E}[Y(1) - Y(0) \mid 1, \text{Post}]}_{\text{ATT}} + \underbrace{\{\Delta \mathbb{E}[Y(0) \mid 1] - \Delta \mathbb{E}[Y(0) \mid 0]\}}_{\text{non-parallel-trends bias}}$$

### 5. Read. If the bias equals zero (*parallel trends*), then $\widehat{\text{DiD}} = \text{ATT}$ .



## The bias term *is* selection bias — on trends, not levels

The “non-parallel-trends bias” isn’t a new concept. It’s selection bias, in a different costume.

$$\text{bias} = \underbrace{\Delta \mathbb{E}[Y(0) \mid D=1]}_{\text{how treated's } Y(0) \text{ moves}} - \underbrace{\Delta \mathbb{E}[Y(0) \mid D=0]}_{\text{how controls' } Y(0) \text{ moves}}$$

- **Cross-section selection bias:** are treated and control different in *levels* of  $Y(0)$ ?
- **DiD’s selection bias:** are treated and control different in *trends* of  $Y(0)$ ?
- Different levels are fine. Unit fixed effects soak them up.
- What can’t differ is the *rate* at which  $Y(0)$  moves across groups.

### The trade DiD makes vs. randomization

RCTs: treated and controls identical in everything — levels and trends.

DiD: identical only in *trends* of  $Y(0)$ . **Cheaper assumption. Weaker identification.**

## Case 1: NA holds, never-treated comparison

**Switching equation.** Treated-post is  $Y(1)$ ; the other three cells are  $Y(0)$ .

$$\widehat{\text{DiD}} = [\mathbb{E}[Y(1)|1, \text{Post}] - \mathbb{E}[Y(0)|1, \text{Pre}]] - [\mathbb{E}[Y(0)|0, \text{Post}] - \mathbb{E}[Y(0)|0, \text{Pre}]]$$

**Add the zero.** Insert  $+\mathbb{E}[Y(0)|1, \text{Post}] - \mathbb{E}[Y(0)|1, \text{Post}]$ . Rearrange:

$$\widehat{\text{DiD}} = \underbrace{\mathbb{E}[Y(1) - Y(0)|1, \text{Post}]}_{\text{ATT}} + \underbrace{\{\Delta\mathbb{E}[Y(0)|1] - \Delta\mathbb{E}[Y(0)|0]\}}_{\text{non-PT bias} = 0 \text{ under PT}}$$

### Numerical example

ATT = 10, PT bias = 0.  $\widehat{\text{DiD}} = 10$ . ✓

## Case 2: NA fails — anticipation contaminates $Y_{Pre}$

Suppose the treated group anticipates and adjusts *before* the policy lands. Their pre-period outcome contains some  $Y(1)$  response:  $\mathbb{E}[Y|1, Pre] = \mathbb{E}[Y(0)|1, Pre] + \text{ATT}_{pre}$ .

Same +0 trick, propagated:

$$\widehat{DiD} = \underbrace{ATT_{Post}}_{\text{what we want}} + \underbrace{\text{non-PT bias}}_{=0} - \underbrace{ATT_{pre}}_{\text{anticipation}}$$

### Numerical example

$$ATT = 10, \text{PT bias} = 0, ATT_{pre} = 3. \quad \widehat{DiD} = 10 - 3 = 7. \text{ biased down}$$

*If treatment effects are roughly constant, DiD can collapse to zero.*

## Case 3: the control group also receives treatment in Post

Now the comparison group's Post outcome is also  $Y(1)$ , not  $Y(0)$ . Switching equation:

$$\widehat{\text{DiD}} = [\mathbb{E}[Y(1)|1, \text{Post}] - \mathbb{E}[Y(0)|1, \text{Pre}]] - [\mathbb{E}[Y(1)|0, \text{Post}] - \mathbb{E}[Y(0)|0, \text{Pre}]]$$

**Add two zeros** — one missing counterfactual on each side — and rearrange:

$$\widehat{\text{DiD}} = \underbrace{\mathbb{E}[Y(1) - Y(0)|1, \text{Post}]}_{\text{ATT}} + \underbrace{\{\Delta\mathbb{E}[Y(0)|1] - \Delta\mathbb{E}[Y(0)|0]\}}_{\text{non-PT bias} = 0 \text{ under PT}} - \underbrace{\mathbb{E}[Y(1) - Y(0)|0, \text{Post}]}_{\text{ATT}_{D=0, \text{Post}} \text{ (spillover)}}$$

### Numerical example

$\text{ATT} = 10$ ,  $\text{PT bias} = 0$ ,  $\text{ATT}_{D=0, \text{Post}} = 4$ .  $\widehat{\text{DiD}} = 10 - 4 = 6$ . **biased toward zero**

## Case 4: the comparison group is *always* treated

The comparison group was treated *before* our data starts. Both their Pre and Post outcomes are  $Y(1)$ , not  $Y(0)$ . Switching equation:

$$\widehat{\text{DiD}} = [\mathbb{E}[Y(1)|1, \text{Post}] - \mathbb{E}[Y(0)|1, \text{Pre}]] - [\mathbb{E}[Y(1)|0, \text{Post}] - \mathbb{E}[Y(1)|0, \text{Pre}]]$$

**Add two zeros** —  $\pm \mathbb{E}[Y(0)|0, \text{Post}]$  and  $\pm \mathbb{E}[Y(0)|0, \text{Pre}]$  on the comparison side — and rearrange:

$$\widehat{\text{DiD}} = \underbrace{\mathbb{E}[Y(1)-Y(0)|1, \text{Post}]}_{\text{ATT}} + \underbrace{\{\Delta \mathbb{E}[Y(0)|1] - \Delta \mathbb{E}[Y(0)|0]\}}_{\text{non-PT bias} = 0 \text{ under PT}} - \underbrace{\Delta \text{ATT}_{D=0}}_{\text{change in comparison TE}}$$

### Numerical example

$\text{ATT} = 10$ ,  $\text{PT bias} = 0$ ,  $\Delta \text{ATT}_{D=0} = 2$ .  $\widehat{\text{DiD}} = 10 - 2 = 8$ . **biased toward zero**

*Four cases, four failures of the punchline. The +0 trick still works — only the zero you add changes.*

## Summary: four ways the $2 \times 2$ decomposes

| What holds                       | $\widehat{DiD} =$        |
|----------------------------------|--------------------------|
| NA, never-treated control        | $ATT + PT\text{-bias}$   |
| PT, no NA, never-treated control | $ATT_{Post} - ATT_{Pre}$ |
| PT, NA, always-treated control   | $ATT - \Delta ATT_{D=0}$ |
| PT, NA, SUTVA fails (spillover)  | $ATT - ATT_{D=1, Post}$  |

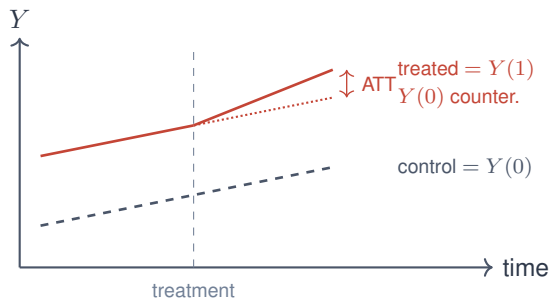
*All four derivable the same way: write down the  $2 \times 2$ , apply the switching equation, add the right series of zeros.*

LESSON

# 5. Card and Krueger: Parallel Trends, Visualized

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## Parallel trends, drawn



$$\mathbb{E}[Y(0) \mid D=1, \text{Post}] - \mathbb{E}[Y(0) \mid D=1, \text{Pre}] = \mathbb{E}[Y(0) \mid D=0, \text{Post}] - \mathbb{E}[Y(0) \mid D=0, \text{Pre}]$$

*A  $2 \times 2$  of  $Y(0)$  terms. **If it isn't four  $Y(0)$ s, it isn't the PTA.***



## Under PT, the 2×2 hits the ATT

With (i) **No Anticipation**, (ii) **Parallel Trends in  $Y(0)$** , and (iii) a clean (never-treated) comparison group:

$$\text{ATT}_{\text{post}} = \underbrace{\mathbb{E}[\Delta Y \mid D = 1]}_{\text{treated change}} - \underbrace{\mathbb{E}[\Delta Y \mid D = 0]}_{\text{control change}}$$

- The 2×2 calculation equals the population's ATT.
- Four averages. Three subtractions. One target parameter.

**This is the punchline. Everything that follows is variations on this theme.**

## New Jersey raises its minimum wage. Pennsylvania doesn't.

- **April 1992**: New Jersey raises its minimum wage from \$4.25 to \$5.05 — a 19% increase.
- Pennsylvania, just over the river, stays at \$4.25.
- Card and Krueger survey 410 fast-food restaurants on both sides of the border, before (Feb 1992) and after (Nov 1992).
- The natural  $2 \times 2$ : **treated = NJ, control = PA, pre/post the policy.**

*Card & Krueger, American Economic Review, 1994.*

## What Card and Krueger *found*

| Employment per restaurant | Pre (Feb 1992) | Post (Nov 1992) | Change                       |
|---------------------------|----------------|-----------------|------------------------------|
| NJ (treated)              | 20.44          | 21.03           | +0.59                        |
| PA (control)              | 23.33          | 21.17           | -2.16                        |
| DiD (NJ – PA)             |                |                 | <b>+2.76 jobs/restaurant</b> |

- Employment in NJ fast food *rose* slightly, relative to PA, after the wage hike.
- Standard error puts the effect roughly indistinguishable from zero — but the sign is the opposite of what theory predicted.
- **No measurable disemployment effect.** The textbook had failed a real test.

## The reaction was personal

*No self-respecting economist would claim that increases in the minimum wage increase employment. Such a claim becomes equivalent to a denial that there is even minimal scientific content in economics.*

— James Buchanan (Nobel 1986), *WSJ*, April 1996

*It cost me a lot of friends. People that I had known for many years became very angry or disappointed. They thought that in publishing our work we were being traitors to the cause of economics as a whole.*

— David Card, interview

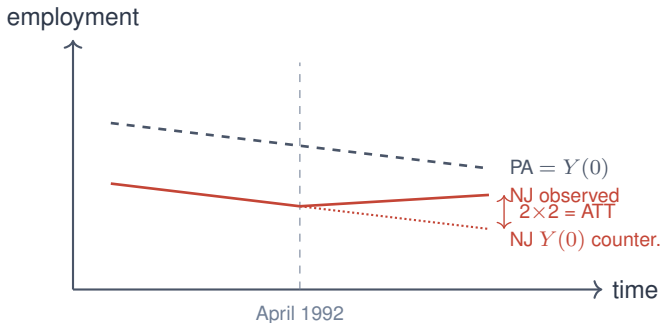
*The price of a credible empirical paper that contradicted the consensus. Krueger paid more.*

## Hear it from them — three short interview clips

- **Orley Ashenfelter on the profession's reaction** (start 31:20)  
[youtu.be/M0tbuRX4eyQ?t=1882](https://youtu.be/M0tbuRX4eyQ?t=1882)
- **Card on the hostility from peers** (1:07–1:10)  
[youtu.be/1soLdywFb\\_Q?t=4020](https://youtu.be/1soLdywFb_Q?t=4020)
- **Card on what happened to Alan Krueger** (1:16–1:17)  
[youtu.be/1soLdywFb\\_Q?t=4556](https://youtu.be/1soLdywFb_Q?t=4556)

*The DiD literature did not arrive politely. It was earned through paper-by-paper combat, much of it personal.*

## Parallel trends visualized: the control *is* the counterfactual



*The dotted line is what NJ would have done absent the policy. We never see it.*

***PT lets PA's slope stand in for it.***

**Aside: there's also an *across-group* PT we never name**

## **The usual statement (within-group, across-time)**

The comparison group's  $Y(0)$  trend stands in for what the treated group's  $Y(0)$  trend would have been.

## **The same assumption, stated from the other side (across-group, within-time)**

The treated group's  $Y(0)$  trend (had they not been treated) would equal the comparison group's  $Y(0)$  trend.

- Two phrasings of one assumption:  $Y(0)$  trends are common across groups.
- Event-study  $\beta_k$  vs  $t = -1$  computes the within-group across-time piece.
- Regression 3 ( $\Delta Y$  on  $D$ ) and Regression 4 ( $D_t$  on Post) are FWL mirrors — same PT from the two directions.

*ASIDE — drop in or drop out if behind schedule.*

## Pre-trends are the gut check — not a proof of PT

- Card–Krueger held up because in the pre-period, NJ and PA fast-food employment moved together.
- If NJ employment had already been diverging before April 1992, the +2.76 estimate would have been mostly trend, not policy.
- Flat pre-trends → evidence (not proof) the design holds.
- **The 2×2 was credible because of the pre-period parallelism — not because of the regression coefficient.**

*More later (Roth, Roth–Sant’Anna): when pre-trends look flat but the test still over-rejects.*



LESSON

# 6. Open the Sheet

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## Open the Sheet: stability and trends, by hand

### Live demo

Compute the four cells, three subtractions, and the by-hand event study — in a spreadsheet.

[docs.google.com/spreadsheets/d/1onabpc14JdrGo6NFv0zCWo-nuWDLLV2L1qNogDT9SBw](https://docs.google.com/spreadsheets/d/1onabpc14JdrGo6NFv0zCWo-nuWDLLV2L1qNogDT9SBw)

*Four tabs in sequence: Potential Outcomes  $\rightarrow$  2X2  $\rightarrow$  Balancing  $\rightarrow$  2XT #1.*

LESSON

# 7. Standard Errors

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## Three flavors of standard error

- **Vanilla (homoskedastic)**: assumes  $\text{Var}(\varepsilon_i \mid X_i) = \sigma^2$  for all  $i$ .
- **Robust (Eicker–Huber–White, HC)**: each residual speaks for itself — handles heteroskedasticity.
- **Cluster-robust (CRVE)**: groups residuals by cluster; within-cluster errors correlate freely.

*Same point estimate, three different SEs. The SE you pick is a claim about the error structure.*

# HC1 numerically: each $\hat{e}_i^2$ weighted by leverage

**Six observations.** Three pairs at  $\tilde{x} \in \{-2, 0, +2\}$ . Fitted  $\hat{\beta}_1 = 1$ ; residuals shown.

| $i$    | cluster $g$ | $\tilde{x}_i$ | $\hat{e}_i$ | $\hat{e}_i^2$ | $\tilde{x}_i^2 \cdot \hat{e}_i^2$ |
|--------|-------------|---------------|-------------|---------------|-----------------------------------|
| 1      | 1           | -2            | +0.5        | 0.25          | 1.00                              |
| 2      | 1           | -2            | +0.5        | 0.25          | 1.00                              |
| 3      | 2           | 0             | -1.0        | 1.00          | 0.00                              |
| 4      | 2           | 0             | -1.0        | 1.00          | 0.00                              |
| 5      | 3           | +2            | +0.5        | 0.25          | 1.00                              |
| 6      | 3           | +2            | +0.5        | 0.25          | 1.00                              |
| meat = |             |               |             |               | 4.00                              |

With  $n = 6$ ,  $k = 2$ , and  $S_{xx} = \sum_i \tilde{x}_i^2 = (-2)^2 + (-2)^2 + 0 + 0 + 2^2 + 2^2 = 16$ :

$$\widehat{\text{Var}}_{\text{HC1}}(\hat{\beta}_1) = \frac{n}{n-k} \cdot \underbrace{\frac{1}{S_{xx}}}_{\text{bread}} \cdot \underbrace{4.00}_{\text{meat}} \cdot \underbrace{\frac{1}{S_{xx}}}_{\text{bread}} = \frac{6}{4} \cdot \frac{4.00}{16 \cdot 16} = \frac{6}{4} \cdot \frac{4.00}{256} = 0.02344$$

$$\Rightarrow \text{SE}_{\text{HC1}} = \sqrt{0.02344} \approx 0.153$$

Obs 3–4 contribute zero because  $\tilde{x} = 0$  kills the leverage weight. **HC1 treats all 6 obs as independent.**

## Why HC-robust barely helps in DiD

The HC correction reweights residuals by  $X_i X_i'$ . In a DiD with binary  $D_i \in \{0, 1\}$  and roughly balanced groups:

- Treated and control units have similar *leverage*: each side carries  $\approx$  half the weight in  $X'X$ .
- Residual variance is roughly symmetric across the two groups (binary  $D$  doesn't generate a fan shape in  $\hat{e}^2$ ).
- The weighted sum  $\sum_i \hat{e}_i^2 X_i X_i'$  ends up not very different from  $\hat{\sigma}^2(X'X)$ .

### Consequence

$SE_{HC} \approx SE_{vanilla}$  for typical DiD specifications.

**The over-rejection problem is not heteroskedasticity. It's something else.**

## CRVE numerically: same 6 obs, group scores, then square

**Same 6 observations.** Now compute  $s_g = \sum_{i \in g} \tilde{x}_i \hat{e}_i$  for each cluster, then  $s_g^2$ :

| $i$ | $g$ | $\tilde{x}_i$ | $\hat{e}_i$ | $\tilde{x}_i \hat{e}_i$ |        |       |         |
|-----|-----|---------------|-------------|-------------------------|--------|-------|---------|
|     |     |               |             |                         | $g$    | $s_g$ | $s_g^2$ |
| 1   | 1   | -2            | +0.5        | -1                      |        |       |         |
| 2   | 1   | -2            | +0.5        | -1                      | 1      | -2    | 4       |
| 3   | 2   | 0             | -1.0        | 0                       | 2      | 0     | 0       |
| 4   | 2   | 0             | -1.0        | 0                       | 3      | +2    | 4       |
| 5   | 3   | +2            | +0.5        | +1                      | meat = |       | 8       |
| 6   | 3   | +2            | +0.5        | +1                      |        |       |         |

With  $G = 3$ ,  $n = 6$ ,  $k = 2$ , and  $S_{xx} = \sum_i \tilde{x}_i^2 = 16$ , so  $S_{xx}^2 = 256$ :

$$\widehat{\text{Var}}_{\text{CRVE}}(\hat{\beta}_1) = \frac{G}{G-1} \cdot \frac{n-1}{n-k} \cdot \frac{\text{meat}}{S_{xx}^2} = \frac{3}{2} \cdot \frac{5}{4} \cdot \frac{8}{256} = 0.05859 \Rightarrow \text{SE}_{\text{CRVE}} \approx 0.242$$

**HC1:** 0.153 vs. **CRVE:** 0.242 (**58% larger**). Same data. The cluster cross-terms changed the *meat* from 4 to 8.

## Bertrand, Duflo & Mullainathan (QJE 2004): the placebo-law test

### “How Much Should We Trust Differences-in-Differences Estimates?”

- BDM generate *placebo laws*: assign random treatment dates to states in a real CPS panel.
- True effect by construction:  $\beta = 0$ . So  $\Pr(\text{reject } H_0)$  should equal  $\alpha = 5\%$ .
- **Vanilla SE**: rejection rate  $\approx 45\%$ . Wildly over-rejects.
- **Heteroskedasticity-robust (HC1)**: rejection rate *also*  $\approx 45\%$ . **HC barely moves the needle.**
- **Cluster-robust at state level**: rejection rate  $\approx 5\%$ . The fix.

### Why HC doesn't save you

Binary, balanced  $D$  + within-state serial correlation. The het-correction has nothing to do with the actual problem. **Clustering is doing all the work.**



LESSON

# 8. Population Weighting

---

## Ten people, two counties

| Person | $Y(0)$ | $Y(1)$ | $\delta_i$ | County |
|--------|--------|--------|------------|--------|
| Alan   | 0      | 1      | 1          | 1      |
| Betty  | 1      | 1      | 0          | 1      |
| Chad   | 1      | 1      | 0          | 1      |
| Daniel | 0      | 0      | 0          | 1      |
| Edith  | 0      | 1      | 1          | 1      |
| Frank  | 0      | 1      | 1          | 1      |
| George | 0      | 0      | 0          | 1      |
| Hank   | 0      | 1      | 1          | 1      |
| Ida    | 1      | 0      | -1         | 2      |
| Janet  | 1      | 0      | -1         | 2      |

*County 1: eight people. County 2: two people. Compute the average  $\delta$  for each county, and across all ten. Same answer?*

## Same data, two weights, two parameters

- **Within counties:** County 1 avg  $\delta = \frac{1}{8}(1+0+0+0+1+1+0+1) = +0.5$ .  
County 2 avg  $\delta = \frac{1}{2}(-1+(-1)) = -1.0$ .
- **ATE for the average person:**  $\frac{1}{10}(8 \cdot 0.5 + 2 \cdot (-1)) = +0.2$ .
- **ATE for the average county:**  $\frac{1}{2}(0.5 + (-1)) = -0.25$ .

*Same potential outcomes. **Different signs.***

*Whichever you report depends on what you're trying to answer.*

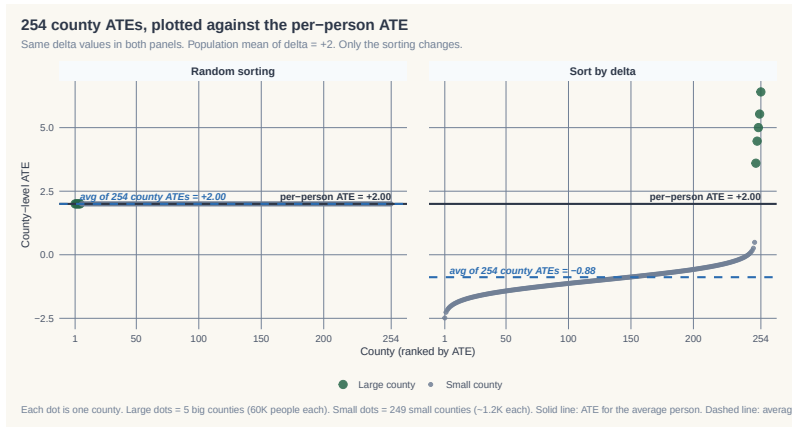
## Two parameters, two answers

Scale the example up: **Texas, 254 counties, 30M people**. 5 biggest counties hold roughly half the population. Heterogeneous  $\delta_i$  across people, with the high- $\delta$  folks concentrated in the 5 big counties and the low- $\delta$  folks in the 249 small ones.

- **Per-person ATE:**  $\frac{1}{N} \sum_i \delta_i$  — weight every person by  $1/N$ . **Answer: +2.**
- **Avg of county ATEs:**  $\frac{1}{254} \sum_c \bar{\delta}_c$  — weight every county by  $1/254$ . **Answer: -0.88.**

*Same data. Two weights. Two parameters. Sometimes two signs.*

# Sorting on $\delta$ can flip the sign of your answer



*Random: every county's ATE equals the population's.  $\delta$ -sort: per-person line at +2, avg of county ATEs flips to  $-0.88$ .*

*Random sorting is the benchmark — rarely reality. People choose where to live, and on traits that move  $\delta$ .*

## The weighted mean is a covariance correction

For any outcome  $Y$  and any weight  $S$ :

$$\tilde{\mathbb{E}}[Y] \equiv \frac{\mathbb{E}[SY]}{\mathbb{E}[S]} = \mathbb{E}[Y] + \frac{\text{Cov}(S, Y)}{\mathbb{E}[S]}$$

- Weighted mean = unweighted mean + a **covariance correction**.
- $\text{Cov}(S, Y) > 0 \Rightarrow$  weighted exceeds unweighted;  $< 0 \Rightarrow$  smaller.
- Ten-people table: weight by county size  $S \in \{8, 2\}$  and compute  $\tilde{\mathbb{E}}[\delta]$ .

### Stata

```
gen size = 8 if county==1
replace size = 2 if county==2
summarize delta [aweight=size]
* weighted mean = +0.2
summarize delta
* unweighted mean = -0.25
```

### R

```
df$size <- ifelse(df$county==1, 8, 2)
weighted.mean(df$delta, df$size)
# weighted mean = +0.2
mean(df$delta)
# unweighted mean = -0.25
```

## Pick a target parameter

- Three weights. Three signs. Three answers.
- The choice is about *which causal effect you actually want*.
- **Person-weighted** for aggregate human welfare: how many people are affected, by how much?
- **Unit-weighted** when locations have autonomy: each policy unit chooses independently — the unit *is* the agent.

## Today's target: the ATT.

*The average effect for the units that actually received the treatment.*

LESSON

# 9. Event Studies: PT Made Falsifiable

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## Event studies are PT falsifications. What is a falsification?

- Popperian reasoning: you don't confirm a theory — you try to falsify it.
- A rival theory earns scrutiny by making a *prediction you can check*.
- Claim a confounder. Derive what *else* it would do. Test that.
- If the prediction holds → you haven't confirmed the rival, but you haven't ruled it out either.
- If the prediction fails → the rival weakens.

*Popperian asymmetry. Failing to reject the rival does not confirm you. But rejecting the rival's prediction is real evidence.*

## Two forms of falsification

- **Same outcome, different group.** A group your treatment shouldn't reach, but the rival would.  
*Min wage on low-wage employment? Re-run on higher-wage workers.*
- **Same group, different outcome.** An outcome your treatment shouldn't move, but the rival would.  
*Cigarette tax on smoking-related hospitalizations? Re-run on appendicitis and fractures.*

*PT falsifications come in both forms. Event studies are one (same group, different “outcome” — a period when no treatment happened).*

## Famous falsification 1: obesity contagion

- Christakis & Fowler (2007): **obesity is contagious** in social networks.
- Cohen-Cole & Fletcher (2008): re-run the same model on **height, acne, headaches** — same Framingham network, same identification.
- All three also show “contagion.”

*We have not been able to become taller by having tall friends.*

— Cohen-Cole & Fletcher (2008)

*Same group, different outcome. The result isn't necessarily wrong — the empirical model is.*

## Famous falsification 2: milk addiction

- Becker–Murphy (1988): serial correlation in cigarette / alcohol  $\Rightarrow$  “**rational addiction.**”
- Auld & Grootendorst (2004, *J. Health Econ.*): apply the same estimator to **milk, eggs, oranges.**
- All three look “rationally addictive.”

*Same group, different outcome. Nobody is rationally addicted to milk. **The identification — not the goods — is what's broken.***

*Point: PT makes predictions beyond the event study. Placebo groups and placebo outcomes are the other two forms.*

## Parallel trends is about $Y(0)$ , not “trends” abstractly

The parallel-trends assumption is a statement about an unobserved object:

$$\mathbb{E}[Y(0)_t - Y(0)_{t-1} \mid D=1] = \mathbb{E}[Y(0)_t - Y(0)_{t-1} \mid D=0]$$

- PT is the counterfactual untreated outcome's trend. **Not** a statement about OBSERVED  $Y$  trends in general.
- Observed  $Y$  for treated post-treatment is  $Y(1)$ , not  $Y(0)$ . The PT assumption can't be read off it.
- The confusion comes from saying “trends are parallel.” What we mean is “**untreated potential outcome trends are parallel.**”

### Why this matters

We never see treated  $Y(0)$  post-treatment. Pre-treatment data is the only window onto whether a similar assumption *could* hold post. That's where event studies come in.

## Event studies: pre-treatment placebos for PT

**The falsification logic.** Suppose some confounder both shifts  $Y(0)$  trends *and* is associated with treatment timing. That's a PT violation.

- If a similar confounder structure existed *before* treatment, we should see treated and control  $Y$  trends diverge in the pre-period too.
- Estimate placebo “effects” at  $t = -2, -3, \dots, -q$  where there is no treatment. If those coefficients are large or trending, we've rejected the assumption that drives the post-period.

### Popperian asymmetry

Flat pre-trends *fail to reject* PT — they do not prove it.

Sloped pre-trends *reject* the design.

**Pre-treatment placebos are a falsification, not a proof.**

## An event study is the $2 \times 2$ viewed in motion

- Instead of one pre-period and one post-period, use *many*.
- Estimate a separate effect at each event-time relative to treatment.
- Plot them:  $\hat{\beta}_{-q}, \dots, \hat{\beta}_{-2}, \hat{\beta}_{-1}, \hat{\beta}_0, \hat{\beta}_{+1}, \dots, \hat{\beta}_{+q}$ .
- The graph tells two stories: **pre-trends** (left of zero) and **dynamics** (right of zero).

*Same arithmetic as the  $2 \times 2$ , repeated against a common reference period.*

## Every coefficient is a $2 \times 2$ against $t = -1$

For each event-time  $k$  (with  $t = -1$  as the reference):

$$\hat{\beta}_k = (\bar{Y}_k^T - \bar{Y}_{-1}^T) - (\bar{Y}_k^C - \bar{Y}_{-1}^C)$$

- Two long differences, one subtraction — the  $2 \times 2$  again, with  $t = -1$  as the pre-period.
- Pre-period  $\hat{\beta}_k$  ( $k < -1$ ): asks whether trends were already diverging.
- Post-period  $\hat{\beta}_k$  ( $k \geq 0$ ): the dynamic ATTs.

*Drop  $k = -1$  as the reference. Multicollinearity won't let you keep all the dummies.*



## The saturated OLS specification

$$Y_{it} = \alpha_i + \tau_t + \sum_{t \neq t_0-1} \beta_t \cdot (D_i \cdot \mathbf{1}\{\text{year} = t\}) + \varepsilon_{it}$$

- $\alpha_i$  — unit fixed effects (absorb levels).
- $\tau_t$  — calendar-year fixed effects (absorb common shocks).
- One interaction per calendar year, omitting  $t_0 - 1$  (the year just before treatment) as the reference.
- Each  $\beta_t$  is the  $2 \times 2$  of the previous slide: (treated change from  $t_0-1$  to  $t$ ) — (control change from  $t_0-1$  to  $t$ ).

*In a  $2 \times T$  design the calendar year is the event-time. We save  $k$  for staggered treatment, where cohorts have different treatment dates.*

# Computing $\hat{\beta}_{2004}$ on castle.dta — by hand vs. OLS

Castle doctrine cohort treated in 2006. Use 2005 as the omitted reference. The 2004 coefficient is one event-study  $2 \times 2$ .

## By hand

```
su l_homicide if treat==1 & year==2004
* mean = 1.802504
su l_homicide if treat==1 & year==2005
* mean = 1.775398
su l_homicide if treat==0 & year==2004
* mean = 1.254353
su l_homicide if treat==0 & year==2005
* mean = 1.282884

di (1.802504-1.775398) - (1.254353-1.282884)
* 0.05564
```

$$\hat{\beta}_{2004} = \mathbf{0.05564}$$

## Via OLS in Stata

```
reg l_homicide ///
    treat##ib2005.year, ///
    cluster(state)

di _b[1.treat#2004.year]
* 0.05564
```

$$\hat{\beta}_{2004} = \mathbf{0.05564}$$

*Cluster-robust SE: 0.060. Match to 5 decimals.*

*Every event-study coefficient is a  $2 \times 2$  relative to  $t = -1$  (here, 2005).*

LESSON

# 10. Five Pieces of a Strong DiD

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## Medicaid expansion and near-elderly mortality

Miller, Johnson & Wherry (2021), *Quarterly Journal of Economics*.

- Did the ACA Medicaid expansion (2014) reduce mortality among the **near-elderly** (ages 55–64)?
- Comparison: states that expanded vs. states that didn't, before vs. after 2014.
- Headline finding:  $\approx 9.2\%$  **reduction** in near-elderly mortality.

### Why *near-elderly*?

Everyone 65+ has Medicare — Medicaid expansion is irrelevant for them. The elderly become a natural **placebo group** living in the same economies.

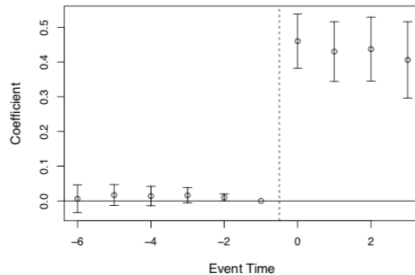
*Today: walk through how they make the case using the five pieces of a strong DiD.*

## Five pieces of a strong DiD — start with bite

1. **Bite** — the policy actually changed what it was supposed to change.
2. **Event study** — pre/post coefficients around treatment.
3. **Falsification** — run a rival theory on a placebo group or outcome.
4. **Main result** — the DiD on the actual outcome.
5. **Mechanism** — the causal pathway from  $D$  to  $Y$ .

*If the policy didn't actually move what it was supposed to move, nothing downstream is attributable to it.*

## Bite #1: Medicaid take-up jumped in expansion states

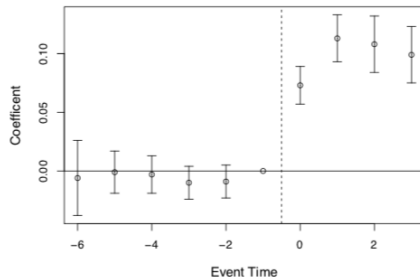


(a) Medicaid Eligibility

**Did the policy actually move people onto Medicaid?** Yes — enrollment rose sharply in expansion states post-2014, flat in non-expansion.

Without bite, there's no story. The mandate has to actually do something before we can attribute downstream changes to it.

## Bite #2: any-insurance coverage rose

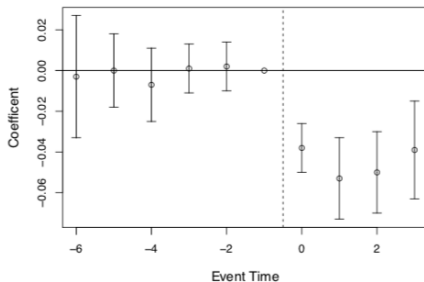


(b) Medicaid Coverage

***Did the new Medicaid enrollees actually gain coverage they didn't have?*** Or did they just shift from private to Medicaid?

*Any-insurance coverage rose — the expansion brought in the previously uninsured, not the already-insured.*

### Bite #3: the uninsured rate fell



(c) Uninsured

**The mirror image of Bite #2.** Uninsured rate dropped sharply in expansion states post-2014. Three measures (Medicaid up, any-insurance up, uninsured down) converge on the same conclusion: **the policy bit.**



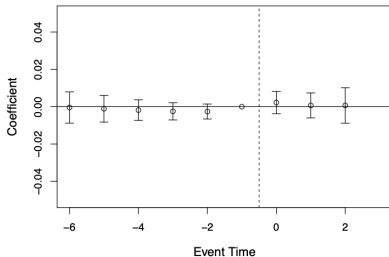
## Five pieces of a strong DiD — now falsify

1. **Bite** — the policy moved what it was supposed to move. (*established.*)
2. **Event study** — pre/post coefficients around treatment.
3. **Falsification** — run a rival theory on a placebo group or outcome and try to reject *it*.
4. **Main result** — the DiD on the actual outcome.
5. **Mechanism** — the causal pathway from  $D$  to  $Y$ .

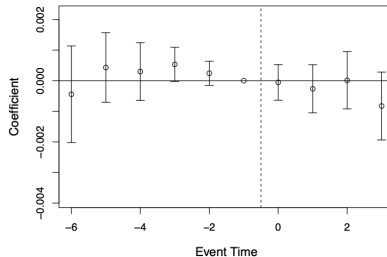
*The rival: “some 2014 shock hit expansion states.” The falsification: run the same model on a group the policy shouldn’t reach.*

## Falsification: same model, on the 65+ population — no effect

*Age 65+ in 2014*



(c) Medicaid Coverage



(d) Annual Mortality

**Same states, same years, same model — but run on the 65+ population.** *They all have Medicare, so Medicaid expansion shouldn't touch their mortality.*

*A rival theory ("some 2014 shock hit expansion states") would predict a mortality effect here. It didn't appear.*

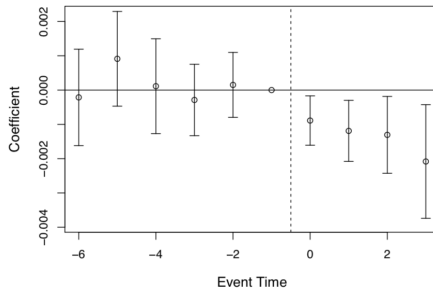
**Rival rejected.**

## Five pieces of a strong DiD — the event study and main result

1. **Bite** — established.
2. **Event study** — pre/post coefficients around treatment.
3. **Falsification** — rival rejected.
4. **Main result** — the DiD on the actual outcome.
5. **Mechanism** — the causal pathway from  $D$  to  $Y$ .

*Now we look at the outcome that matters: mortality. The event study is the main result, viewed in motion.*

## The main result: $\approx 9.2\%$ reduction in near-elderly mortality



- Post-2014: mortality declines in expansion states.  $\approx 9.2\%$  **reduction**.
- Pre-period dots near zero, but **CIs are wide** — you cannot reject a straight line trending downward (*Rambachan & Roth 2024* sensitivity).
- The other four pieces are why the editors believed it. A critic now has to explain bite *and* falsification *and* the dynamic post-trend — not just one thing.

## Five pieces of a strong DiD — and the fifth: mechanism

1. **Bite** — established.
2. **Event study** — established.
3. **Falsification** — rival rejected.
4. **Main result** — 9.2% reduction in near-elderly mortality.
5. **Mechanism** — the causal pathway from  $D$  to  $Y$ .
  - Mechanisms are the part of a result that's **transportable** — they extend across contexts where the same pathway operates.
  - Reduced-form top-line results often combine *several* mechanisms, which makes interpretation harder.
  - Here: disease-related deaths fall and compound, consistent with treated chronic conditions.

## The five pieces, all five present in Medicaid

---

|                      |                                                                                      |
|----------------------|--------------------------------------------------------------------------------------|
| <b>Bite</b>          | Three figures: Medicaid take-up up, any-insurance up, uninsured down                 |
| <b>Event study</b>   | Pre-trends flat; post-trends decline in expansion states                             |
| <b>Falsification</b> | No effect on elderly (65+) — same states, same years, same model                     |
| <b>Main result</b>   | $\approx 9.2\%$ reduction in mortality among the near-elderly                        |
| <b>Mechanism</b>     | Disease-related deaths fall and compound; consistent with treated chronic conditions |

---

*None of these alone is the case. **Together they are powerful.***

# What we've done today — and tomorrow

## Today:

1. DiD's empirical roots — Semmelweis, Ashenfelter.
2. DiD = calculation + assumption: the  $2 \times 2$  and parallel trends.
3. Five regressions, one number.
4. Decomposing the  $2 \times 2$  — the +0 trick and three failure cases.
5. Card and Krueger: parallel trends visualized.
6. Standard errors — vanilla, robust, cluster.
7. Population weighting: pick a target parameter.
8. Event studies as falsifications, by hand and via OLS.

**Tomorrow:** when the  $2 \times 2$  stops working — covariates, staggered adoption (Goodman-Bacon, Callaway–Sant'Anna, Sun–Abraham, BJS, dCdH).